STUDENT LESSON OUTLINE · YEAR 9-10

Advanced Water Safety

Unit Overview & Lesson Outline

This outline shows you what you will learn, how you will be assessed, and the key vocabulary for each lesson across the full unit.

Year 9–10 HPE | 5 Lessons | Victorian Curriculum 2.0

VC2HP10P02 · VC2HP10P08 · VC2HP10P09 · VC2HP10M01

📋 Unit Overview

BY THE END OF THIS UNIT YOU WILL KNOW

- Victorian & global drowning rates and how to calculate them
- The physiology of drowning and how CPR keeps the brain alive
- Social determinants that influence who drowns and why
- Which prevention strategies have the strongest evidence
- How Australian approaches compare to international ones

BY THE END YOU WILL UNDERSTAND

- Why data alone doesn't prevent drowning
- How structural factors create unequal drowning risk
- Why some prevention strategies work and others don't
- The link between community action and individual safety
- How to evaluate health information critically

BY THE END YOU WILL BE ABLE TO DO

- Calculate and interpret drowning rate statistics
- Evaluate prevention evidence and justify decisions
- Compare international water safety approaches
- Design an evidence-based health communication product
- Argue a health policy position using real data

Prerequisite knowledge: This unit builds directly on the Year 7–8 Swim & Water Safety unit. You should already know: beach flags and their meanings, the DRS ABCD emergency action plan, the 5 LSV SwimSafe rules, rip current escape strategies, and basic Victorian drowning statistics. If you haven't completed Year 7–8, speak with your teacher for a catch-up summary.

1 Advanced Data Analysis — Victorian & Global Drowning Trends

⏱ 50 min | Workbook Section 1 | VC2HP10P02 · VC2HP10P08 | Individual + Pairs

LEARNING INTENTION

We are learning to **calculate, compare, and critically interpret** Victorian and global drowning statistics to understand who is most at risk and why.

SUCCESS CRITERIA

- [ ] I can calculate Victorian drowning rates per 100,000 population and compare them to the national average, showing all working. `VC2HP10P08`
- [ ] I can analyse CALD over-representation in drowning data and explain at least two structural reasons for the pattern — not just individual ones. `VC2HP10P02`
- [ ] I can create a correctly labelled bar graph from real LSV data and write a two-sentence interpretation of the trend it shows. `VC2HP10P08`

KEY VOCABULARY

- **drowning rate** — deaths per 100,000 population
- **CALD** — Culturally & Linguistically Diverse
- **over-representation** — higher % than population share
- **non-fatal drowning** — near-drowning, survived
- **unpatrolled** — no lifeguard present
- **trend** — pattern over time in data
- **demographic** — statistical group characteristic
- **disproportion** — unequal distribution

KEY ACTIVITY

Calculate Victoria's drowning rate, compare to national and global averages, and build a data visualisation from the LSV Drowning Report 2023–24.

END-OF-LESSON REFLECTION

Exit ticket: In one sentence — what is the single most important data insight from today, and why does it matter?

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2 Physiology of Drowning & CPR Mechanics

⏱ 50 min | Workbook Section 2 | VC2HP10P08 · VC2HP10M01 | Individual

LEARNING INTENTION

We are learning to **explain the physiological sequence of drowning** and justify why early bystander CPR is the single most important factor in survival outcomes.

SUCCESS CRITERIA

- [ ] I can describe the correct sequence of physiological events during drowning using all five required terms: submersion, hypoxia, laryngospasm, cardiac arrest, cerebral hypoxia. `VC2HP10P08`
- [ ] I can explain in plain language why starting CPR within 4 minutes produces significantly better outcomes than starting after 8 minutes, using the terms cardiac output and oxygenated blood. `VC2HP10M01`
- [ ] I can explain what secondary drowning is, why it is clinically significant, and what action should be taken after any submersion incident — even if the person appears fine. `VC2HP10P08`

KEY VOCABULARY

hypoxia
oxygen deprivation at tissue level

laryngospasm
involuntary airway closure

cardiac arrest
heart stops pumping effectively

cerebral hypoxia
oxygen deprivation to the brain

cardiac output
volume of blood heart pumps per minute

CPR
30 compressions : 2 breaths, 100–120 bpm

secondary drowning
delayed pulmonary oedema post-submersion

1-10-1 rule
cold water survival framework

KEY ACTIVITY

Sequence the physiological events of drowning in order, then write a plain-language explanation of the 4-minute CPR window.

END-OF-LESSON REFLECTION

Explain the 4-minute window to someone who has never heard of it — use plain language, no medical jargon.

3 Social Determinants of Drowning Risk

⏱ 50 min | Workbook Section 3 | VC2HP10P02 · VC2HP10P08 | Pairs + Jigsaw

LEARNING INTENTION

We are learning to **analyse how social, cultural, and structural factors** influence who drowns and why — moving beyond individual explanations to systemic ones.

SUCCESS CRITERIA

- [ ] I can identify at least three social determinants of drowning risk and explain the specific mechanism by which each one increases risk — not just that it does. `VC2HP10P02`
- [ ] I can distinguish between individual explanations (personal behaviour) and structural explanations (systemic factors) for drowning risk, using real Victorian data as evidence. `VC2HP10P02`
- [ ] I can propose three specific, practical actions a Victorian school could take to reduce CALD drowning disproportionality — with enough detail to actually implement them. `VC2HP10P08`

KEY VOCABULARY

social determinants
non-medical factors shaping health

structural factors
systemic, not individual, causes

masculinity norms
social expectations for males

socioeconomic status
income, education, occupation level

health inequity
unfair, avoidable health differences

intersectionality
overlapping social identities

community capacity
ability of a group to address issues

correlation
relationship between two variables

KEY ACTIVITY

Jigsaw activity — each pair analyses one social determinant and reports back to the class, building a shared picture of the factors that drive Victorian drowning disproportionality.

END-OF-LESSON REFLECTION

Name ONE social determinant a school can address and ONE it cannot. Why is that distinction important for planning prevention strategies?

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4 Prevention Strategies & International Comparison

⏱ 50 min | Workbook Sections 4–5 | VC2HP10P08 · VC2HP10P09 | Individual + Research

LEARNING INTENTION

We are learning to **evaluate drowning prevention strategies against evidence** and compare Australian approaches to those used internationally, identifying what Victoria could learn or adopt.

SUCCESS CRITERIA

- [ ] I can evaluate five prevention strategies (pool fencing, swim lessons, lifeguard patrols, warning signs, CALD programs) against the criteria of evidence strength, target population, and key limitations. `VC2HP10P08`
- [ ] I can compare the water safety approaches of RLSSA (Australia), RLSS UK, and the American Red Cross, identifying at least one substantive difference in their core messages or strategies. `VC2HP10P09`
- [ ] I can argue, with evidence cited from at least one primary source with a URL, whether a specific international approach should or should not be adopted in Victoria. `VC2HP10P09`

KEY VOCABULARY

evidence hierarchy
ranking of research quality

primary source
original data or research

pool fencing
4-sided isolation fencing

patrolled location
supervised by lifeguard

Float to Live
RLSS UK primary survival message

RLSSA
Royal Life Saving Society Australia

intervention
deliberate action to change outcomes

success metric
measurable indicator of effectiveness

RESEARCH SITES

royallifesaving.com.au · rlss.org.uk/education · redcross.org/take-a-class/water-safety · lsv.com.au · who.int

END-OF-LESSON REFLECTION

Which ONE prevention strategy has the strongest evidence? Which has the weakest? Defend your answer.

5 Diving, Spinal Injuries & High-Risk Teenage Behaviour

⏱ 50 min | Workbook Section 6 | VC2HP10P02 · VC2HP10P08 | Individual + Discussion

LEARNING INTENTION

We are learning to **analyse Australian data on diving-related spinal injury** — identifying the specific risk factors, population groups, and locations driving this pattern — and to evaluate prevention messages using an evidence-based framework.

SUCCESS CRITERIA

- [ ] I can explain, using AIHW data, why unsafe diving accounts for 10% of spinal cord injuries and 20% of quadriplegia cases in Australia — and identify the highest-risk age group. `VC2HP10P08`
- [ ] I can analyse three recent Australian case studies (Terrigal, Torquay, Bawley Beach) and identify the specific mechanism — not just the behaviour — that led to injury in each case. `VC2HP10P02`
- [ ] I can construct a structural explanation for why 95% of aquatic spinal injury patients are male, using the social determinants framework from Section 3. `VC2HP10P02`
- [ ] I can evaluate the "Feet First, First Time" prevention message against evidence and propose one specific complementary strategy for a named high-risk group. `VC2HP10P08`

KEY VOCABULARY

cervical spine · quadriplegia · spinal cord injury · sandbank · tidal depth variation · false confidence · inland waterway · remote rescue delay · sediment · aquatic spinal injury · feet first first time · AIHW · RLSSA · diving-related injury

KEY DATA — KEEP THIS HANDY

- Unsafe diving = 10% of all spinal cord injuries and 20% of quadriplegia cases in Australia (AIHW)
- 95% of aquatic spinal injury patients are male; 75%+ are under 30; 66% suffer permanent disability
- Highest incidence by age group: children aged 10–14
- RLSSA: "More people drown in inland waterways than any other location in Australia"
- +34% increase in drowning among 15–20 year olds post-COVID vs pre-COVID (RLSSA)

END-OF-LESSON REFLECTION

The data shows the highest incidence of diving spinal injury is in children aged 10–14. What specific social and environmental factors explain why this age group is most at risk — and what would a targeted prevention response look like?

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LEARNING INTENTION

We are learning to **synthesise our unit knowledge into an evidence-based communication product** — a policy brief, campaign, or school audit — that argues a clear position and proposes measurable outcomes.

SUCCESS CRITERIA

- [ ] I can produce a coherent draft of my chosen assessment option that includes at least three credible sources with URLs and at least one accurate Victorian or global statistic. `VC2HP10P09`
- [ ] I can clearly identify my target audience or addressee and demonstrate that my communication product is designed specifically for them — not as a generic response. `VC2HP10P09`
- [ ] I can use the assessment checklist to peer-review a classmate's draft and provide two specific, constructive written suggestions for improvement. `VC2HP10P08`

ASSESSMENT OPTIONS — CHOOSE ONE

Option A — Policy Brief
600–800 words to Victorian Minister for Sport. Two recommendations + evidence + success metrics.

Option B — Campaign Product
Target ONE high-risk group. Campaign name, message, visual, platform + 100-word justification.

Option C — School Audit ★ Advanced
Audit your school's provision. Benchmark against Curriculum 2.0. Priority matrix + recommendations.

UNIT REFLECTION — COMPLETE AFTER LESSON 5

What is the single most important thing I learned in this unit that I didn't know before?

What question does this unit leave me wanting to investigate further?

🎯 Assessment Overview — What Is Expected

All options must include:
✓ At least 3 sources with URLs
✓ At least 1 Victorian or global statistic

A-standard work:
Goes beyond description. Analyses, evaluates, and argues with evidence. Acknowledges counter-

Key resources to cite:
lsv.com.au/research
royallifesaving.com.au
who.int/drowning
victoriancurriculum.vcaa.vic.edu.au

✓ Clear target audience / addressee
✓ Specific call to action or recommendation
✓ Connection to a social determinant or physiological concept

arguments. Proposes measurable outcomes. Cites primary sources. Is written specifically for the named audience.

rlss.org.uk/education
redcross.org/water-safety